

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Collected and processed **biological samples**, conducting **laboratory analysis** on fresh and conserved forage, concentrate feedstuffs, and animal tissue samples.
- Recorded experimental information accurately and orderly, making optimal use of **automated recording systems** for research data management.
- Assisted researchers in commissioning new equipment and techniques, actively participating in the development and adoption of **new research methodologies**.
- Provided technical support for a range of existing laboratory assays, ensuring adherence to **experimental protocols** and quality standards.
- Cooperated with management to ensure health, safety, and welfare in the laboratory environment, adhering strictly to the **Safety Statement**.
- Managed data handling and computer operations for laboratory experiments, including **database management** and statistical analysis of results.
- Collaborated with a **dynamic team** of researchers, technologists, and farm staff to optimize animal performance in pasture-based production systems.

SKILLS

Laboratory Analysis – Feed analysis, Forage analysis, Animal tissue analysis, Sample processing, Laboratory assays

Data Management – Database management, Statistical analysis, Data handling, Automated recording systems

Research Support – Experimental protocols, New techniques, Equipment commissioning, Stakeholder collaboration

Health & Safety – Safety Statement compliance, Risk management, Laboratory procedures

General Computing – Computer skills, Microsoft Office Suite, Data recording

PROJECTS

Feed Analysis Pipeline Development (HCL Technologies)

- Developed Python scripts to automate the processing of raw laboratory data from feed analysis instruments, reducing manual data entry time by approximately 40%.
- Implemented data validation checks within the scripts to ensure accuracy and consistency of results before input into the main analysis database.
- Created scheduled reports summarizing key feedstuff metrics for research teams, improving their access to critical experimental data.

Laboratory Sample Tracking System (MSc Data Analytics Project)

- Designed and implemented a relational database schema using SQL to track biological samples from collection through to analysis and reporting.
- Built interactive dashboards in **Power BI** to visualize sample throughput, analysis status, and identify potential bottlenecks in the laboratory workflow.
- Applied **DAX** measures to calculate turnaround times for different sample types, providing insights for process optimization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)